

Swapnith Varma

Portfolio Website

swapnithvarma0@gmail.com | +91-9573756869 | [linkedin.com/in/swapnith-varma-ba86b4163](https://www.linkedin.com/in/swapnith-varma-ba86b4163)

🎓 EDUCATION

• **B.Tech, Electronics and Communication Engineering**, JNTUH — CGPA: 9.05/10 2016 – 2020

👤 SUMMARY

Software Developer with nearly 5 years of experience building and optimizing enterprise CAD-PLM integration systems using C++, C#, and SQL. Strong CAD domain knowledge with hands-on experience in Creo Toolkit APIs and developing shared Toolkit frameworks used across multiple CAD addins. Proficient in object-oriented design and SOLID principles, with a track record of delivering significant performance improvements (70–80%). Experienced in Agile/Scrum workflows and actively upskilling in LLM internals and transformer architecture, leveraging AI tools like Amazon Q daily to accelerate and optimize development.

✂ SKILLS

- **Programming Languages:** C++, C#, SQL, Baan 4GL, JavaScript, Python
- **Frameworks/Systems:** .NET, Infor LN, React, Node.js, REST APIs
- **Tools & IDEs:** Git, SVN, JIRA, Amazon Q, Docker, Eclipse, Visual Studio
- **Learning/Exposure:** LLMs, NLP Preprocessing, Attention Mechanisms
- **Cloud Platforms:** Infor LN Cloud (Enterprise SaaS – CAD–PLM integration)

🏢 EXPERIENCE

• Infor

Software Engineer (Jun 2024 – Present), Associate Software Engineer (Apr 2021 – May 2024)

Hyderabad, India

Enhancements in Office Integration to PLM Cloud

C#

- Built end-to-end integration between Infor PLM Cloud and Microsoft 365 using Microsoft APIs, enabling seamless saving of Word, Excel, and PowerPoint files directly from OneDrive to PLM Cloud.
- Developed a batch registration pipeline to bulk upload and manage large volumes of Office files via Microsoft APIs, significantly reducing manual effort for enterprise customers.
- Enabled multiple customers to centralize and manage their Office files in PLM Cloud, helping teams manage documents more efficiently across their organization.

Custom Defined Fields From CAD Integration

C++, C#, SQL, 4GL

- Designed a logic-driven solution to import ERP-defined custom fields into CAD Integration, enabling design-specific customizations and seamless ERP–CAD synchronization, widely used by all customers.

Performance Optimization in CAD-PLM Integration

C++, C#, SQL, 4GL

- Optimized CAD–PLM integration save operations by enhancing Toolkit and server-side logic, achieving 70–80% faster saves while ensuring reliable and stable data processing at scale in PLM Cloud.
- Designed and implemented JSON-based request handling to manage multiple interlinked database save operations and applied concurrency concepts such as multi-threading, synchronization, and request orchestration.

Creo Integration

C++

- Migrated Creo integration from v8 to v9+ by adapting to new API structures, rewriting functionalities, and validating across CAD operations, ensuring continued compatibility for enterprise customers.
- Developed top-level download functionality that replicates the full item hierarchy defined in PLM into Creo as a structured assembly, triggered by a single item ID.
- Implemented insert-replace functionality enabling users to swap an existing component in a Creo assembly with a different one fetched from PLM, maintaining assembly structure and relationships.

CAD Design Variant Handling in PLM Cloud

C++, C#, SQL, 4GL

- Enhanced Creo–PLM integration by implementing variant-aware saving functionality using STL that creates separate documents for each design variant with linked neutral files, while maintaining synchronized relationships with the generic document.
- This was a customer-critical requirement for go-live ensuring on-time implementation by overcoming major technical challenges.

Integration of Vertex, Oracle AutoVue Viewer with Infor PLM Cloud *C#, SQL, 4GL, Javascript, Java*

- Integrated Vertex CAD Viewer with PLM Cloud using REST APIs, designing a C# proof-of-concept for file uploads and retrievals, then implementing production integration in Infor LN for 2D/3D previews, markups, and seamless synchronization.
- Integrated Oracle AutoVue Desktop Viewer with PLM Cloud for file download, markup creation, and saving back via a custom button-based interface.
- Delivered seamless CAD visualization and review for on-premise and cloud customers, now actively used in production environments.

Multi Language Mapping in CAD-PLM Cloud Integration *C++, C#, SQL, 4GL*

- Implemented multi-language mapping where users could define Mapping rules in multiple languages which will save data across multiple languages in a single click.
- Actively used by customers, saving time by eliminating the need to switch languages, thereby improving efficiency.

CAD Utility from Creo *C++, C#, SQL, 4GL*

- Extended the CAD Utility to support Creo integration, enabling it to automatically retrieve records, open CAD files, generate neutral files, perform save operations, and close without any user intervention — fully automating background workflows end to end.

Export Import of Mapping Templates across Different Tenants *Infor SQL, 4GL*

- Developed a feature to export CAD Mapping Templates with all associated rules and linked tables, enabling seamless one-click import into other tenants. Eliminated the need for lengthy data migration processes, significantly reducing setup time and effort for customers across multiple environments.
- Built and shipped major Creo integration features across different CAD file operations and PLM synchronization, while contributing to shared Toolkit framework enhancements utilized across Inventor, SolidWorks, and AutoCAD addins, resolving complex defects through in-depth debugging and root cause analysis to ensure timely customer go-lives.
- Worked in **Agile** methodology with monthly sprints and bi-annual releases, collaborating with cross-functional teams to gather requirements, track progress via Jira, and deliver customer-focused features within tight timelines.
- Assisted newly joined and existing team members in resolving issues and clarifying requirements, promoting smoother collaboration and knowledge sharing.
- Actively integrating AI tools like **Amazon Q** into day-to-day development workflows, while building foundational knowledge in **LLM** architecture — having completed data preprocessing pipelines and currently working through attention mechanisms and transformer internals

PROJECTS

Lightweight Software Updater & Release Manager *C#, GIT*

- Built a C#-based light weight auto-update system using GitHub Releases and metadata-driven version tracking, enabling fully automated patch delivery, secure ZIP extraction, and integration DLL deployment for versioned software updates.

Portfolio Website [Live Site](#) *JavaScript, React, CSS, Vite*

- Built a responsive portfolio website using React, JavaScript, and CSS. Implemented dynamic project sections with modal interactions and smooth animations, following a reusable component-based architecture with efficient state management. Optimized performance using Vite and ensured cross-device compatibility.